Site-to-Site IPSec VPN Troubleshoot:

Troubleshooting VPN using debug commands. Start with the debug crypto isakmp command and walk through a successful ISAKMP SA creation. After issue, the clear crypto session command and ping a host from one side to the other side.

R3# Debug crypto isakmp

```
*Oct 26 13:14:55.271: ISAKMP (0): received packet from 192.168.12.1 dport 500 sport 500 Global (N) NEW SA
*Oct 26 13:14:55.271: ISAKMP: Created a peer struct for 192.168.12.1, peer port 500
*Oct 26 13:14:55.271: ISAKMP: New peer created peer = 0x686CBB6C peer_handle = 0x80000002
*Oct 26 13:14:55.275: ISAKMP: Locking peer struct 0x686CBB6C, refcount 1 for crypto_isakmp_process_block
*Oct 26 13:14:55.275: ISAKMP: local port 500, remote port 500
*Oct 26 13:14:55.279: ISAKMP:(0):insert sa successfully sa = 686CB1A8
*Oct 26 13:14:55.287: ISAKMP:(0):Input = IKE_MESG_FROM_PEER, IKE_MM_EXCH
*Oct 26 13:14:55.291: ISAKMP:(0):Old State = IKE_READY  New State = IKE_R_MM1   First packet in the IKE Phase I

*Oct 26 13:14:55.295: ISAKMP:(0): processing SA payload. message ID = 0
*Oct 26 13:14:55.299: ISAKMP:(0): processing vendor id payload
*Oct 26 13:14:55.299: ISAKMP:(0): vendor ID seems Unity/DPD but major 69 mismatch
*Oct 26 13:14:55.299: ISAKMP (0): vendor ID is NAT-T RFC 3947
*Oct 26 13:14:55.299: ISAKMP:(0): pr
R3#ocessing vendor id payload
*Oct 26 13:14:55.303: ISAKMP:(0): vendor ID seems Unity/DPD but major 245 mismatch
*Oct 26 13:14:55.303: ISAKMP (0): vendor ID is NAT-T v7
*Oct 26 13:14:55.303: ISAKMP:(0): processing vendor id payload
*Oct 26 13:14:55.303: ISAKMP:(0): vendor ID seems Unity/DPD but major 157 mismatch
*Oct 26 13:14:55.307: ISAKMP:(0): vendor ID is NAT-T v3
*Oct 26 13:14:55.307: ISAKMP:(0): processing vendor id payload
*Oct 26 13:14:55.307: ISAKMP:(0): vendor ID seems Unity/DPD but major 123 mismatch
*Oct 26 13:14:55.307: ISAKMP:(0): vendor ID is NAT-T v2
*Oct 26 13:14:55.311: ISAKMP:(0):found peer pre-shared key matching 192.168.12.1
*Oct 26 13:14:55.311: ISAKMP:(0): local preshared key found
                                                      peers have exchanged their IKE Phase 1 policies
*Oct 26 13:14:55.311: ISAKMP : Scanning profiles for xauth ...
*Oct 26 13:14:55.311: ISAKMP:(0):Checking ISAKMP transform 1 against priority 5 policy
*Oct 26 13:14:55.311: ISAKMP:      encryption 3DES-CBC
*Oct 26 13:14:55.315: ISAKMP:      hash SHA
R3#AKMP:      default group 2
*Oct 26 13:14:55.315: ISAKMP:      auth pre-share
                                                      Policy matched and accepted
*Oct 26 13:14:55.315: ISAKMP:      life type in seconds
*Oct 26 13:14:55.315: ISAKMP:      life duration (VPI) of  0x0 0x1 0x51 0x80
*Oct 26 13:14:55.319: ISAKMP:(0):atts are acceptable. Next payload is 0
*Oct 26 13:14:55.319: ISAKMP:(0):Acceptable atts:actual life: 0
*Oct 26 13:14:55.319: ISAKMP:(0):Acceptable atts:life: 0              Default Lifetime
*Oct 26 13:14:55.323: ISAKMP:(0):Fill atts in sa vpi_length:4
*Oct 26 13:14:55.323: ISAKMP:(0):Fill atts in sa life_in_seconds:86400
*Oct 26 13:14:55.323: ISAKMP:(0):Returning Actual lifetime: 86400
*Oct 26 13:14:55.323: ISAKMP:(0)::Started lifetime timer: 86400.

*Oct 26 13:14:55.327: ISAKMP:(0): processing vendor id payload
*Oct 26 13:14:55.327: ISAKMP:(0): vendor ID seems Unity/DPD but major 69 mismatch
*Oct 26 13:14:55.327: ISAKMP (0): vendor ID is NAT-T RFC 3947
*Oct 26 13:14:55.327: ISAKMP:(0): processing vendor id payload
*Oct 26 13:14:55.331: ISAKMP:(0): vendor ID
R3#seems Unity/DPD but major 245 mismatch
*Oct 26 13:14:55.331: ISAKMP (0): vendor ID is NAT-T v7
*Oct 26 13:14:55.331: ISAKMP:(0): processing vendor id payload
*Oct 26 13:14:55.331: ISAKMP:(0): vendor ID seems Unity/DPD but major 157 mismatch
*Oct 26 13:14:55.335: ISAKMP:(0): vendor ID is NAT-T v3
*Oct 26 13:14:55.335: ISAKMP:(0): processing vendor id payload
*Oct 26 13:14:55.335: ISAKMP:(0): vendor ID seems Unity/DPD but major 123 mismatch
*Oct 26 13:14:55.339: ISAKMP:(0): vendor ID is NAT-T v2
*Oct 26 13:14:55.339: ISAKMP:(0):Input = IKE_MESG_INTERNAL, IKE_PROCESS_MAIN_MODE
*Oct 26 13:14:55.339: ISAKMP:(0):Old State = IKE_R_MM1  New State = IKE_R_MM1

*Oct 26 13:14:55.347: ISAKMP:(0): constructed NAT-T vendor-rfc3947 ID
```

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*Oct 26 13:14:55.351: ISAKMP:(0): sending packet to 192.168.12.1 my_port 500 peer_port 500 (R) MM_SA_SETUP
*Oct 26 13:14:55.351: ISAKMP:(0):Sending an IKE IPv4 Packet.
*Oct 26 13:14:55.351: ISAKMP:(0):Input = IKE_MESG_INTERNAL, IKE_PROCESS_COMPLETE
*Oct
R3# 26 13:14:55.355: ISAKMP:(0):Old State = IKE_R_MM1  New State = IKE_R_MM2

*Oct 26 13:14:55.455: ISAKMP (0): received packet from 192.168.12.1 dport 500 sport 500 Global (R) MM_SA_SETUP
*Oct 26 13:14:55.455: ISAKMP:(0):Input = IKE_MESG_FROM_PEER, IKE_MM_EXCH
*Oct 26 13:14:55.459: ISAKMP:(0):Old State = IKE_R_MM2  New State = IKE_R_MM3        Received response from the peer

*Oct 26 13:14:55.463: ISAKMP:(0): processing KE payload. message ID = 0
*Oct 26 13:14:55.563: ISAKMP:(0): processing NONCE payload. message ID = 0
*Oct 26 13:14:55.567: ISAKMP:(0):found peer pre-shared key matching 192.168.12.1
*Oct 26 13:14:55.571: ISAKMP:(1001): processing vendor id payload
*Oct 26 13:14:55.571: ISAKMP:(1001): vendor ID is DPD
*Oct 26 13:14:55.575: ISAKMP:(1001): processing vendor id payload
*Oct 26 13:14:55.575: ISAKMP:(1001): speaking to another IOS box!
*Oct 26 13:14:55.575: ISAKMP:(1001): processing vendor id payload          Third and fourth packets  exchanged
*Oct 26 13:14:55.575: ISAKMP:(1001): vendor ID seems Unity/DPD but major 255 mismatch
*Oct 26 13:1
R3#4:55.579: ISAKMP:(1001): vendor ID is XAUTH
*Oct 26 13:14:55.579: ISAKMP:received payload type 20          No NAT device in the way
*Oct 26 13:14:55.579: ISAKMP (1001): His hash no match - this node outside NAT
*Oct 26 13:14:55.579: ISAKMP:received payload type 20
*Oct 26 13:14:55.579: ISAKMP (1001): No NAT Found for self or peer
*Oct 26 13:14:55.583: ISAKMP:(1001):Input = IKE_MESG_INTERNAL, IKE_PROCESS_MAIN_MODE
*Oct 26 13:14:55.583: ISAKMP:(1001):Old State = IKE_R_MM3  New State = IKE_R_MM3

*Oct 26 13:14:55.591: ISAKMP:(1001): sending packet to 192.168.12.1 my_port 500 peer_port 500 (R) MM_KEY_EXCH
*Oct 26 13:14:55.591: ISAKMP:(1001):Sending an IKE IPv4 Packet.

*Oct 26 13:14:55.595: ISAKMP:(1001):Input = IKE_MESG_INTERNAL, IKE_PROCESS_COMPLETE
*Oct 26 13:14:55.595: ISAKMP:(1001):Old State = IKE_R_MM3  New State = IKE_R_MM4

*Oct 26 13:14:55.687: ISAKMP (1001): received packet from 192.168.12.1 dport 500 sport 500 Global (R) MM_KEY_EXCH
*Oct 26 13:14:55.691: ISAKMP:(1001):Input = IKE_MESG_FROM_PEER,
R3#IKE_MM_EXCH
*Oct 26 13:14:55.691: ISAKMP:(1001):Old State = IKE_R_MM4  New State = IKE_R_MM5        Fifth packet has been sent

*Oct 26 13:14:55.699: ISAKMP:(1001): processing ID payload. message ID = 0
*Oct 26 13:14:55.699: ISAKMP (1001): ID payload
        next-payload : 8
        type         : 1
        address      : 192.168.12.1
        protocol     : 17
        port         : 500
        length       : 12
*Oct 26 13:14:55.703: ISAKMP:(0):: peer matches *none* of the profiles
*Oct 26 13:14:55.703: ISAKMP:(1001): processing HASH payload. message ID = 0
*Oct 26 13:14:55.707: ISAKMP:(1001): processing NOTIFY INITIAL_CONTACT protocol 1
        spi 0, message ID = 0, sa = 686CB1A8
*Oct 26 13:14:55.707: ISAKMP:(1001):SA authentication status:        Peer successfully authenticated phase I
        authenticated
*Oct 26 13:14:55.707: ISAKMP:(1001):SA has been authenticated with 192.168.12.1
*Oct 26 13:14:55.711: ISAKMP:(1001):SA authentication status:
        authenticated
*Oct 26 13:14:55.711: ISAKMP:(1001): Process initial contact,
bring down existing phase 1 and 2 SA's with local 192.168.23.3 remot
R3#e 192.168.12.1 remote port 500
*Oct 26 13:14:55.715: ISAKMP: Trying to insert a peer 192.168.23.3/192.168.12.1/500/,  and inserted successfully 686CBB6C
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.
*Oct 26 13:14:55.715: ISAKMP:(1001):Input = IKE_MESG_INTERNAL, IKE_PROCESS_MAIN_MODE
*Oct 26 13:14:55.715: ISAKMP:(1001):Old State = IKE_R_MM5  New State = IKE_R_MM5

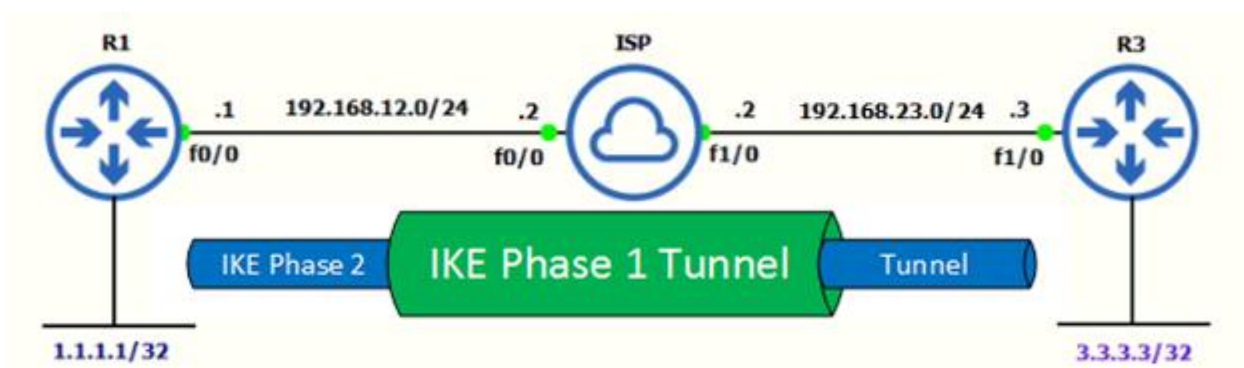
*Oct 26 13:14:55.727: ISAKMP:(1001):SA is doing pre-shared key authentication using id type ID_IPV4_ADDR
*Oct 26 13:14:55.727: ISAKMP (1001): ID payload
        next-payload : 8
        type         : 1
        address      : 192.168.23.3
        protocol     : 17
        port         : 500
        length       : 12
*Oct 26 13:14:55.731: ISAKMP:(1001):Total payload length: 12
*Oct 26 13:14:55.735: ISAKMP:(1001): sending packet to 192.168.12.1 my_port 500 peer_port 500 (R) MM_KEY_EXCH
*Oct 26 13:14:55.735: ISAKMP:(1001):Sending an IKE IPv4 Packet.
*Oct 26 13:14:55.739: ISAKMP:(1001):Input = IKE_MESG_INTERNAL, IKE_PROCESS_COMPLETE
*Oct 26 13:14:55.739: ISAKMP:(1001):Old State = IKE_R_MM5  N
R3#ew State = IKE_P1_COMPLETE

*Oct 26 13:14:55.747: ISAKMP:(1001):Input = IKE_MESG_INTERNAL, IKE_PHASE1_COMPLETE
*Oct 26 13:14:55.747: ISAKMP:(1001):Old State = IKE_P1_COMPLETE  New State = IKE_P1_COMPLETE

*Oct 26 13:14:55.791: ISAKMP (1001): received packet from 192.168.12.1 dport 500 sport 500 Global (R) QM_IDLE
*Oct 26 13:14:55.791: ISAKMP: set new node 1647259992 to QM_IDLE
*Oct 26 13:14:55.795: ISAKMP:(1001): processing HASH payload. message ID = 1647259992
*Oct 26 13:14:55.795: ISAKMP:(1001): processing SA payload. message ID = 1647259992
*Oct 26 13:14:55.799: ISAKMP:(1001):Checking IPSec proposal 1
*Oct 26 13:14:55.799: ISAKMP: transform 1, ESP_AES
*Oct 26 13:14:55.799: ISAKMP:   attributes in transform:
*Oct 26 13:14:55.799: ISAKMP:      encaps is 1 (Tunnel)
*Oct 26 13:14:55.799: ISAKMP:      SA life type in seconds

*Oct 26 13:14:55.799: ISAKMP:      SA life duration (basic) of 3600
*Oct 26 13:14:55.803: ISAKMP:      SA life type in kilobytes
*Oct
R3# 26 13:14:55.803: ISAKMP:      SA life duration (VPI) of  0x0 0x46 0x50 0x0
*Oct 26 13:14:55.803: ISAKMP:      authenticator is HMAC-MD5
*Oct 26 13:14:55.807: ISAKMP:      key length is 128                    Accepts the transform-set
*Oct 26 13:14:55.807: ISAKMP:(1001):atts are acceptable.
*Oct 26 13:14:55.807: ISAKMP:(1001): processing NONCE payload. message ID = 1647259992
*Oct 26 13:14:55.811: ISAKMP:(1001): processing ID payload. message ID = 1647259992
*Oct 26 13:14:55.811: ISAKMP:(1001): processing ID payload. message ID = 1647259992
*Oct 26 13:14:55.823: ISAKMP:(1001):QM Responder gets spi
*Oct 26 13:14:55.827: ISAKMP:(1001):Node 1647259992, Input = IKE_MESG_FROM_PEER, IKE_QM_EXCH
*Oct 26 13:14:55.827: ISAKMP:(1001):Old State = IKE_QM_READY  New State = IKE_QM_SPI_STARVE
*Oct 26 13:14:55.835: ISAKMP:(1001): Creating IPSec SAs
*Oct 26 13:14:55.839:         inbound SA from 192.168.12.1 to 192.168.23.3 (f/i)  0/ 0
        (proxy 1.1.1.1 to 3.3.3.3)
*Oct 26 13:14:55.839:         has spi 0xD8A24D94 and conn_id 0
R3#
*Oct 26 13:14:55.839:         lifetime of 3600 seconds
*Oct 26 13:14:55.839:         lifetime of 4608000 kilobytes
*Oct 26 13:14:55.843:         outbound SA from 192.168.23.3 to 192.168.12.1 (f/i) 0/0
        (proxy 3.3.3.3 to 1.1.1.1)
*Oct 26 13:14:55.843:         has spi  0xA5DAA9ED and conn_id 0
*Oct 26 13:14:55.843:         lifetime of 3600 seconds
*Oct 26 13:14:55.843:         lifetime of 4608000 kilobytes
*Oct 26 13:14:55.847: ISAKMP:(1001): sending packet to 192.168.12.1 my_port 500 peer_port 500 (R) QM_IDLE
*Oct 26 13:14:55.847: ISAKMP:(1001):Sending an IKE IPv4 Packet.
*Oct 26 13:14:55.851: ISAKMP:(1001):Node 1647259992, Input = IKE_MESG_INTERNAL, IKE_GOT_SPI
*Oct 26 13:14:55.851: ISAKMP:(1001):Old State = IKE_QM_SPI_STARVE  New State = IKE_QM_R_QM2
*Oct 26 13:14:55.963: ISAKMP (1001): received packet from 192.168.12.1 dport 500 sport 500 Global (R) QM_IDLE

*Oct 26 13:14:55.967: ISAKMP:(1001):deleting node 1647259992 error FALSE reason "QM done (awai
R3#t)"                                                        Phase II quick mode is complete
*Oct 26 13:14:55.967: ISAKMP:(1001):Node 1647259992, Input = IKE_MESG_FROM_PEER, IKE_QM_EXCH
*Oct 26 13:14:55.971: ISAKMP:(1001):Old State = IKE_QM_R_QM2  New State = IKE_QM_PHASE2_COMPLETE
```



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Troubleshooting Phase 1:

Let us Make some changes to the ISAKMP policy on the remote peer and clear the crypto session by issuing the clear crypto session command. When do the debug after we clear the session, the changes I made should be reflected.

Mismatch Encryption in the ISAKMP Policy:

R3(config)#crypto isakmp policy 5
R3(config-isakmp)#authentication pre-share
R3(config-isakmp)#encryption aes
R3(config-isakmp)#group 2
R3(config-isakmp)#hash sha
R3(config-isakmp)#lifetime 86400
R3(config-isakmp)#exit
R3(config)#crypto isakmp key cisco123 address 192.168.23.3
R3# debug crypto isakmp

```
*Oct 26 10:22:47.959: ISAKMP:(0):Encryption algorithm offered does not match policy!
*Oct 26 10:22:47.959: ISAKMP:(0):atts are not acceptable. Next payload is 0
*Oct 26 10:22:47.959: ISAKMP:(0):no offers accepted!
*Oct 26 10:22:47.959: ISAKMP:(0): phase 1 SA policy not acceptable! (local 192.168.23.3 remote 192.168.12.1)
*Oct 26 10:22:47.963: ISAKMP (0): incrementing error counter on sa, attempt 1 of 5: construct_fail_ag_init
*Oct 26 10:22:47.963: ISAKMP:(0): Failed to construct AG informational message.
*Oct 26 10:22:47.963: ISAKMP:(0): sending packet to 192.168.12.1 my_port 500 peer_port 500 (R) MM_NO_STATE
*Oct 26 10:22:47.967: ISAKMP:(0):Sending an IKE IPv4 Packet.
*Oct 26 10:22:47.967: ISAKMP:(0):peer does not do paranoid keepalives.

*Oct 26 10:22:47.967: I
R3#SAKMP:(0):deleting SA reason "Phase1 SA policy proposal not accepted" state (R) MM_NO_STATE (peer 192.168.12.1)
```

Mismatch Encryption

Mismatch Hash Algorithm in the ISAKMP Policy:

R3(config)#crypto isakmp policy 5
R3(config-isakmp)#authentication pre-share
R3(config-isakmp)#encryption 3des
R3(config-isakmp)#group 2
R3(config-isakmp)#hash md5
R3(config-isakmp)#lifetime 86400
R3(config-isakmp)#exit
R3(config)#crypto isakmp key cisco123 address 192.168.23.3
R3# debug crypto isakmp

```
*Oct 26 10:48:15.551: ISAKMP:      default group 2
*Oct 26 10:48:15.551: ISAKMP:      auth pre-share
*Oct 26 10:48:15.551: ISAKMP:      life type in seconds
*Oct 26 10:48:15.551: ISAKMP:      life duration (VPI) of  0x0 0x1 0x51 0x80
*Oct 26 10:48:15.555: ISAKMP:(0):Hash algorithm offered does not match policy!
*Oct 26 10:48:15.555: ISAKMP:(0):atts are not acceptable. Next payload is 0
*Oct 26 10:48:15.555: ISAKMP:(0):no offers accepted!
*Oct 26 10:48:15.555: ISAKMP:(0): phase 1 SA policy not acceptable! (local 192.168.23.3 remote 192.168.12.1)
```

Mismatch Hash algorithm

Mismatch Diffie-Hellman Group in ISAKMP Policy:

R3(config)#crypto isakmp policy 5
R3(config-isakmp)#authentication pre-share
R3(config-isakmp)#encryption 3des
R3(config-isakmp)#group 5
R3(config-isakmp)#hash sha
R3(config-isakmp)#lifetime 86400
R3(config)#crypto isakmp key cisco123 address 192.168.23.3
R3# debug crypto isakmp

```
R3#SAKMP:      default group 2
*Oct 26 10:51:44.495: ISAKMP:      auth pre-share
*Oct 26 10:51:44.495: ISAKMP:      life type in seconds
*Oct 26 10:51:44.495: ISAKMP:      life duration (VPI) of  0x0 0x1 0x51 0x80
*Oct 26 10:51:44.499: ISAKMP:(0):Diffie-Hellman group offered does not match policy!
*Oct 26 10:51:44.499: ISAKMP:(0):atts are not acceptable. Next payload is 0
*Oct 26 10:51:44.503: ISAKMP:(0):no offers accepted!
*Oct 26 10:51:44.503: ISAKMP:(0): phase 1 SA policy not acceptable! (local 192.168.23.3 remote 192.168.12.1)
```

Mismatch Diffie-Hellman Group

Mismatch Authentication Type in ISAKMP Policy:

R3(config)#crypto isakmp policy 5
R3(config-isakmp)# authentication rsa-sig
R3(config-isakmp)#encryption 3des
R3(config-isakmp)#group 2
R3(config-isakmp)#hash sha
R3(config-isakmp)#lifetime 86400
R3(config)#crypto isakmp key cisco123 address 192.168.23.3
R3# debug crypto isakmp

```
R3#6 11:04:14.311: ISAKMP:      life duration (VPI) of  0x0 0x1 0x51 0x80
*Oct 26 11:04:14.315: ISAKMP:(0):Authentication method offered does not match policy!
*Oct 26 11:04:14.315: ISAKMP:(0):atts are not acceptable. Next payload is 0
*Oct 26 11:04:14.315: ISAKMP:(0):no offers accepted!
*Oct 26 11:04:14.315: ISAKMP:(0): phase 1 SA policy not acceptable! (local 192.168.23.3 remote 192.168.12.1)
*Oct 26 11:04:14.319: ISAKMP (0): incrementing error counter on sa, attempt 1 of 5: construct_fail_ag_init
*Oct 26 11:04:14.319: ISAKMP:(0): Failed to construct AG informational message.
*Oct 26 11:04:14.319: ISAKMP:(0): sending packet to 192.168.12.1 my_port 500 peer_port 500 (R) MM_NO_STATE
```

Mismatch Authentication type

Pre-Shared Keys are wrong:

R3(config)#crypto isakmp key cisco12 address 192.168.23.3
R3# debug crypto isakmp

```
*Oct 26 10:56:22.031: ISAKMP (1002): received packet from 192.168.12.1 dport 500 sport 500 Global (R) MM_KEY_EXCH
R3#t 26 10:56:22.035: ISAKMP: reserved not zero on ID payload!
*Oct 26 10:56:22.035: %CRYPTO-4-IKMP_BAD_MESSAGE: IKE message from 192.168.12.1 failed its sanity check or is malformed
*Oct 26 10:56:22.035: ISAKMP (1002): incrementing error counter on sa, attempt 1 of 5: reset_retransmission
*Oct 26 10:56:23.035: ISAKMP:(1002): retransmitting phase 1 MM_KEY_EXCH...
*Oct 26 10:56:23.035: ISAKMP (1002): incrementing error counter on sa, attempt 2 of 5: retransmit phase 1
*Oct 26 10:56:23.035: ISAKMP:(1002): retransmitting phase 1 MM_KEY_EXCH
*Oct 26 10:56:23.039: ISAKMP:(1002): sending packet to 192.168.12.1 my_port 500 peer_port 500 (R) MM_KEY_EXCH
*Oct 26 10:56:23.039: ISAKMP:(1002):Sending an IKE IPv4 Packet.
```

pre-shared keys wrong

Troubleshooting Phase 2:

Let us jump into Phase 2 troubleshooting. Going to alter IPSec, transform set to let it fail on Phase 2. By changing the transform set, Main Mode exchange complete and Phase 2 start.
I changed the responder transform-set to esp-3des instead of esp-aes like the one the peer is configured to this router and I am using debug crypto ipsec to generate the following output.

Mismatch Transform-Set Attribute:

R1(config)# crypto ipsec transform-set TSET esp-3des esp-md5-hmac
R1(cfg-crypto-trans)#crypto ipsec security-association lifetime seconds 3600
R3#debug crypto ipsec

```
*Oct 26 11:12:40.115: IPSEC(key_engine): got a queue event with 1 KMI message(s)
*Oct 26 11:12:40.171: IPSEC(validate_proposal_request): proposal part #1
*Oct 26 11:12:40.171: IPSEC(validate_proposal_request): proposal part #1,
  (key eng. msg.) INBOUND local= 192.168.23.3, remote= 192.168.12.1,
    local_proxy= 3.3.3.3/255.255.255.255/0/0 (type=1),
    remote_proxy= 1.1.1.1/255.255.255.255/0/0 (type=1),
    protocol= ESP, transform= NONE  (Tunnel),
    lifedur= 0s and 0kb,
    spi= 0x0(0), conn_id= 0, keysize= 128, flags= 0x0
R3#
*Oct 26 11:12:40.175: Crypto mapdb : proxy_match
        src addr     : 3.3.3.3
        dst addr     : 1.1.1.1
        protocol     : 0
        src port     : 0
        dst port     : 0
*Oct 26 11:12:40.179: IPSEC(ipsec_process_proposal): transform proposal not supported for identity:
    {esp-aes esp-md5-hmac }
```

Mismatch or Incompatibility in the transform set

Wrong ACL in the Crypto Map:

Now let us put the wrong ACL in the crypto map or do not put the one that matches traffic for the traffic going from one side to the other. On the responder, it will clearly state that the proxy IDs are what were not supported:

R1(config)#crypto map CMAP 10 ipsec-isakmp
R1(config-crypto-map)#match address VPN
R1(config-crypto-map)#set peer 192.168.23.3
R1(config-crypto-map)#set transform-set TSET
R3#debug crypto ipsec

```
*Oct 26 11:41:46.147: IPSEC(key_engine): got a queue event with 1 KMI message(s)
*Oct 26 11:41:46.207: IPSEC(validate_proposal_request): proposal part #1
*Oct 26 11:41:46.207: IPSEC(validate_proposal_request): proposal part #1,
  (key eng. msg.) INBOUND local= 192.168.12.1, remote= 192.168.23.3,
    local_proxy= 1.1.1.1/255.255.255.255/0/0 (type=1),
    remote_proxy= 3.3.3.3/255.255.255.255/0/0 (type=1),
    protocol= ESP, transform= NONE  (Tunnel),
    lifedur= 0s and 0kb,
    spi= 0x0(0), conn_id= 0, keysize= 128, flags= 0x0
*Oct 26 11:41:46.211: IPSEC(ipsec_process_proposal): proxy identities not supported
```

ACL for IPsec traffic does not match

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Show Commands to Troubleshooting VPN:

This command shows the Internet Security Association Management Protocol (ISAKMP) security associations (SAs) built between peers. This command displays the details for the IKE Phase 1 tunnel including the current status.

```
R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id status
192.168.23.3    192.168.12.1    QM_IDLE           1002 ACTIVE
```

Peer Routers   Phase 1 SA State   Active/Standby

This command, displays the encryption algorithm, hash algorithm, authentication method, and Diffie-Hellman group configured on the device.

```
R1#show crypto isakmp policy

Global IKE policy
Protection suite of priority 5
        encryption algorithm:   Three key triple DES
        hash algorithm:         Secure Hash Standard
        authentication method:  Pre-Shared Key
        Diffie-Hellman group:   #2 (1024 bit)
        lifetime:               86400 seconds, no volume limit

R1#show running-config | sec policy
crypto isakmp policy 5
 encr 3des
 authentication pre-share
 group 2
```

This command will give a quick list of all IKE and IPSec SA sessions. Using the commands can easily verify whether an IPSec tunnel is active, down, or still negotiating.

```
R1#show crypto session
Crypto session current status

Interface: FastEthernet0/0
Session status: UP-ACTIVE
Peer: 192.168.23.3 port 500
  IKE SA: local 192.168.12.1/500 remote 192.168.23.3/500 Active
  IPSEC FLOW: permit ip host 1.1.1.1 host 3.3.3.3
        Active SAs: 2, origin: crypto map
```

This command shows IPsec SAs built between peers. The encrypted tunnel is built between peers for traffic that goes between. Two ESP SAs built inbound and outbound. This command displays the details for the IKE Phase 2 tunnel.

```
R1#show crypto ipsec sa
                                         Crypto MAP name apply under interface
interface: FastEthernet0/0
    Crypto map tag: CMAP, local addr 192.168.12.1    Local Router IP

   protected vrf: (none)                 This is the traffic ready to be encrypted
   local  ident (addr/mask/prot/port): (1.1.1.1/255.255.255.255/0/0)
   remote ident (addr/mask/prot/port): (2.2.2.2/255.255.255.255/0/0)
   current_peer 192.168.22.1 port 500
     PERMIT, flags={origin_is_acl,ipsec_sa_request_sent}
    #pkts encaps: 43, #pkts encrypt: 43, #pkts digest: 43     Encrypted &
    #pkts decaps: 41, #pkts decrypt: 41, #pkts verify: 41     Decrypted Traffic
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 2, #recv errors 0
                                         Local and Remote IPs
     local crypto endpt.: 192.168.12.1, remote crypto endpt.: 192.168.22.1
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
     current outbound spi: 0x60FB37C4(1627076548)
     PFS (Y/N): N, DH group: none

     inbound esp sas:
      spi: 0xC13B9EE6(3241909990)        Transform Set Defined during Configuration
        transform: esp-aes esp-md5-hmac ,
        in use settings ={Tunnel, }      VPN Mode is Tunnel
        conn id: 9, flow_id: SW:9, sibling_flags 80000046, crypto map: CMAP
        sa timing: remaining key lifetime (k/sec): (4500505/3590)
        IV size: 16 bytes
        replay detection support: Y      Anti-Replay Detection is ON
        Status: ACTIVE

  outbound esp sas:
    spi: 0x60FB37C4(1627076548)
      transform: esp-aes esp-md5-hmac ,
      in use settings ={Tunnel, }
      conn id: 10, flow_id: SW:10, sibling_flags 80000046, crypto map: CMAP
      sa timing: remaining key lifetime (k/sec): (4500505/3590)
      IV size: 16 bytes
      replay detection support: Y
      Status: ACTIVE

  outbound ah sas:
```

This command verify and display Phase 1 Preshared Key.

```
R3#show crypto isakmp key
Keyring      Hostname/Address                        Preshared Key

default      192.168.12.1                            cisco123
```

This command displays the details of a crypto map, where it is applied, transform set, ACLs involved, and who the peer address and security association lifetime details.

```
R3#show crypto map
Crypto Map "CMAP" 10 ipsec-isakmp
        Peer = 192.168.12.1
        Extended IP access list VPN-TRAFFIC
            access-list VPN-TRAFFIC permit ip host 3.3.3.3 host 1.1.1.1
        Current peer: 192.168.12.1
        Security association lifetime: 4608000 kilobytes/3600 seconds
        Responder-Only (Y/N): N
        PFS (Y/N): N
        Transform sets={
                TSET:  { esp-3des esp-md5-hmac } ,
        }
        Interfaces using crypto map CMAP:
                FastEthernet1/0
```

Peer IP

Transform Set TSET

Crypto MAP applied

This command verify and check crypto ACLs for hit counts.

```
R1#show ip access-lists
Extended IP access list VPN-TRAFFIC
    10 permit ip host 1.1.1.1 host 3.3.3.3 (14 matches)
```

Verify for mirrored crypto ACLs on each side.

```
R3#show access-lists
Extended IP access list VPN-TRAFFIC
    10 permit ip host 3.3.3.3 host 1.1.1.1 (8 matches)
```